- **Definition:** Given a set of vectors S (possibly infinite), a linear combination is a finite sum of the form

$$\sum_{m=1}^{M} \alpha_m \underline{x}_m \,, \quad \alpha_m \in \mathbb{F} \,, \quad \underline{x}_m \in \mathbb{F}$$

important for vector space lin. comb

- **Definition:** a set of vectors S is linearly independent if $\forall$ finite M and $\{\underline{x}_m\}_{m=1}^{M} \subset S$

$$\sum_{m=1}^{M} \alpha_m \underline{x}_m = 0 \implies \alpha_m = 0 \,, \quad m = 1 \ldots M$$

EX) $S = \left\{ \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\underline{x}_1}, \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\underline{x}_2} \right\}$ $\qquad \alpha_1 \underline{x}_1 + \alpha_2 \underline{x}_2 = \begin{bmatrix} \alpha_2 \\ \alpha_1 \end{bmatrix}$

Note: $\begin{bmatrix} \alpha_2 \\ \alpha_1 \end{bmatrix} = \underline{0} \implies \alpha_1 = \alpha_2 = 0$

- **Definition:** Span of a set of vectors

span(S) = set of all linear combinations of vectors in S

<u>Fact:</u> span(S) is always a subspace (actually the smallest subspace that contains S)

- **Definition:** Basis (Hamel basis, linear vector space basis)

A basis B for a subspace M satisfies:
1) linear independence
2) span$(B) = M$

EX) $S = \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

1) linearly indep. ✓
2)

S is a basis for
- span(S) (always true)
- $\mathbb{R}^2$ if $\mathbb{F} = \mathbb{R}$

- $\mathbb{C}^2$ if $\mathbb{F} = \mathbb{C}$

EX) Is $B = \{ e^{j2\pi n t} \}_{n \in \mathbb{Z}}$ a basis for "well-behaved" $C[0,1]$ functions?

- For each "well-behaved" $f(t) \in C[0,1]$, there exists unique coefficients $\{\alpha_n\}_{n \in \mathbb{Z}}$ s.t.
$f(t) = \sum_{m \in \mathbb{Z}} \alpha_m e^{j2\pi m t}$ ← Fourier Series representation

$\downarrow$

not a finite sum, not a linear combination

$\Downarrow$

B is NOT a (Hamel) basis
(B is a Schauder basis)

- Definition: Dimension

A vector space X is finite-dimensional if there exists a finite, linearly independent set S, s.t. span$\{S\} = X$. The number of elements in S is the dimension of X. If no finite set exists, X is infinite-dimensional!

EX) $\mathbb{R}^N$ and $\mathbb{C}^N$

$S = \{ \underline{e}_1, \underline{e}_2 \dots, \underline{e}_N \}$, $\underline{e}_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$ ← i^{th} position

"standard basis"
"canonical basis" for $\mathbb{R}^N$ or $\mathbb{C}^N$

dimension: N

EX) the set of all real polynomials of degree at most N

finite dimensional ✓
dimension: N+1

$S = \{1, t, t^2, \dots t^N\}$ is a basis

Another basis:

$$S = \left\{ 1, t, \frac{3t^2-1}{2}, \dots \frac{1}{2^N N!} \frac{d^N}{dt^N} (t^2-1)^N \right\}$$

– orthogonal basis, which can be nice

* Why might orthogonality be a good thing for us?

Less sensitive to noise when solving inverse problems!
Allows us to have better-conditioned problems

EX) $C[0,1] \rightarrow$ not a finite dimensional space

EX) $S = \{ x[n] : n \in \mathbb{Z} \} \longrightarrow$ not finite dimensional space

$\Downarrow$

motivates our interest in infinite dimensional cases

Linear Inverse Problems in $\mathbb{C}^N$

Given: $\underline{x} \in \mathbb{C}^N$, $\underbrace{\underline{b}}_{\text{measurements}} \in \mathbb{C}^M$, $\underbrace{\mathcal{A}}_{\text{map}} : \mathbb{C}^N \rightarrow \mathbb{C}^M$, $\mathcal{A}(\underline{x}) = \underline{b}$

Goal: Find $\underline{x}$ given $\mathcal{A}(\cdot)$ and $\underline{b}$

- Definition: a map $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ is a function that assigns to each element $\underline{x} \in \mathcal{X}$ (the domain) a unique element $\underline{y} = \mathcal{A}(\underline{x}) \in \mathcal{Y}$ (the image)

- Definition: linear map satisfies

$$\mathcal{A}(\alpha \underline{x} + \beta \underline{y}) = \alpha \mathcal{A}(\underline{x}) + \beta \mathcal{A}(\underline{y}) \quad \forall \alpha, \beta \in \mathbb{F} \quad \forall \underline{x}, \underline{y} \in \mathcal{X}$$

EX) DSP (convolution btn 2 finite-length signals)

$\underline{x} \in \mathbb{C}^N$ and $\underline{h} \in \mathbb{C}^M$

$\underline{h} * \underline{x} = \mathcal{H}(\underline{x}) : \mathbb{C}^N \to \mathbb{C}^{N+M-1}$

linear map ✓ (LTI systems)

EX) DSP (time reversal)

linear map ✓

EX) truncation (length N signal to length P signal)
$N > P$

linear map ✓

EX) zero-padding (length N signal to length L signal)
$N < L$

linear map ✓

EX) Integer downsampling / upsampling

linear map ✓

EX) DWT, DCT, DFT

linear map ✓

<u>Fact</u>: Any linear map $\mathcal{A}(\cdot)$ from $\mathbb{C}^N$ to $\mathbb{C}^M$ can be written as a matrix:

$$A \in \mathbb{C}^{M \times N}$$

<u>Proof</u>: Any vector $\underline{x} \in \mathbb{C}^N$ can be written as $\underline{x} = \sum_{n=1}^{N} x_n \underline{e}_n$

canonical basis
~ kroniker delta functions

Thus, $\mathcal{A}(\underline{x}) = \mathcal{A}\left(\sum_{n=1}^{N} x_n \underline{e}_n\right)$ linearity

$$= \sum_{n=1}^{N} x_n \mathcal{A}(\underline{e}_n)$$

Define $A \triangleq [A(\underline{e}_1), A(\underline{e}_2), \dots A(\underline{e}_N)]$

we have $\mathcal{A}(\underline{x}) = A\underline{x}$ for $\forall \underline{x} \in \mathbb{C}^N$

- Definition : the range space of $A \in \mathbb{C}^{M \times N}$

$$R(A) \triangleq \{A\underline{x} \in \mathbb{C}^M : \underline{x} \in \mathbb{C}^N\}$$

<u>Facts</u> :
- $R(A)$ is a subspace
- $\text{rank}(A) \triangleq \dim(R(A)) =$ # linearly indep. columns
- $R(A) = \text{span}(\underbrace{\{\underline{a}_1, \underline{a}_2, \cdots \underline{a}_N\}}_{\text{columns of } A})$

- Definition : the nullspace of $A \in \mathbb{C}^{M \times N}$ is

$$\mathcal{N}(A) \triangleq \{\underline{x} \in \mathbb{C}^N : A\underline{x} = \underline{0}\}$$

<u>Facts:</u>
- $\mathcal{N}(A)$ is a subspace
- $\dim(\mathcal{N}(A)) = N - \dim(R(A))$

Solve $A\underline{x} = \underline{b}$ exactly $(A\underline{x} - \underline{b} = \underline{0})$

- Theorem : (Existence) Let $A \in \mathbb{C}^{M \times N}$. The following are equivalent:

① For each $\underline{b} \in \mathbb{C}^M$, there exists <u>at least one</u> solution to $A\underline{x} = \underline{b}$

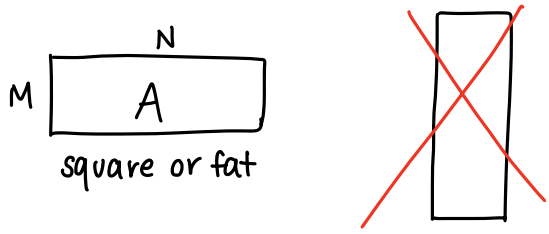
② $R(A) = \mathbb{C}^M$

$\downarrow \dim(R(A))$

③ $\text{rank}(A) = M$

④ The rows of A are linearly independent (full row-rank)

Note: It is necessary (but not sufficient) for $M \leq N$



Proof: $R(A) = \text{span}\{\underline{a}_n\}_{n=1}^{N} \Rightarrow \dim R(A) = \text{rank}(A) \leq N$

$\Rightarrow M \leq N$

EX) consider $A \in \mathbb{C}^{1\times N}$, a solution to $A\underline{x} = \underline{b}$ exists for $\forall b \in \mathbb{C}^1$ unless $A = \underline{0}^H$

EX) $A = \begin{bmatrix} 6 & 5 & 6 & 3 \\ 2 & 3 & 0 & 1 \\ 4 & 4 & 3 & 2 \end{bmatrix} \in \mathbb{C}^{3\times 4}$

- not full row rank $\Rightarrow r_1 + r_2 = 2 \cdot r_3$
- solutions don't exist for $\forall \underline{b} \in \mathbb{C}^3$
- solutions still do exist for certain $\underline{b}$, specifically $b \in R(A)$ $\longrightarrow$ don't give up completely if existance thm. fails

* common misconception: having $M \leq N$ implies that a solution exists. Not always the case.

* we still don't know <u>how</u> to find the solution, but we'll get there soon

- Uniqueness depends on $\mathcal{N}(A)$
  - consider two solutions $\underline{x}_1$ and $\underline{x}_2$ to $A\underline{x} = \underline{b}$
  - $A(\underline{x}_1 - \underline{x}_2) = \underline{0} \Rightarrow \underline{x}_1 - \underline{x}_2 \in \mathcal{N}(A)$
  - If $\mathcal{N}(A) = \{\underline{0}\}$ (trivial null space), then $\underline{x}_1 = \underline{x}_2$ and the solution is

unique

- If $\mathcal{N}(A)$ is non-trivial, but $A\underline{x} = \underline{b}$ have the particular solution $\underline{x}_p$. Then every solution $\underline{x}_0$ satisfies $\underline{x}_0 = \underline{x}_p + \underline{z}$ for some $z \in \mathcal{N}(A)$

- Conversely, for $\forall\ \underline{z} \in \mathcal{N}(A)$, $\underline{x}_p + \underline{z}$ is a solution to $A\underline{x} = \underline{b}$

- the solution set S, $S = \underbrace{\{\underline{x}_p\} + \mathcal{N}(A)}_{\substack{\text{linear variety} \\ \text{(ofset hyperplane/ translated subspace)}}}$

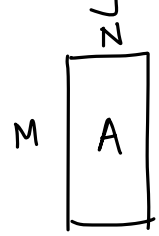
If we have a null-space, there are infinitely many solutions

- Theorem: (Uniqueness)

Let $A \in \mathbb{C}^{M \times N}$. The following are equivalent:

1. If there exists a solution to $A\underline{x} = \underline{b}$, then it is unique
2. $\mathcal{N}(A) = \{\underline{0}\}$ nullspace is trivial
3. $\text{rank}(A) = N = N - \dim(\mathcal{N}(A))$
4. columns of A are linearly independent (full column rank)

<u>Note</u>: It's necessary (but not sufficient) that $N \leq M$



EX) consider $A \in \mathbb{C}^{M \times 1}$. Uniqueness of solutions is guaranteed (if sol.'s exist) unless $A = \underline{0}$

- Theorem : (Existance & Uniqueness)

  Let $A \in \mathbb{C}^{N \times M}$. The following are equivalent :

  1. for each $\underline{b} \in \mathbb{C}^M$, there exists a solution to $A\underline{x} = \underline{b}$ and it is unique
  2. $R(A) = \mathbb{C}^M$ and $\mathcal{N}(A) = \{\underline{0}\}$
  3. $\text{rank}(A) = N = M$
  4. A must be square and non-singular

It will be very rare if all these things are true.

EX) $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \in \mathbb{C}^{3 \times 3}$

not full row/col rank $\Rightarrow$ fails the conditions of the theorem

If we satisfy these conditions, then $\underline{x} = A^{-1} \cdot \underline{b}$ but this is <u>rare</u>.

Generalized Inverses

- Find a matrix B s.t. $\underline{x} = B\underline{b}$ is a solution to $A\underline{x} = \underline{b}$ whenever solutions exist
- Left-Inverses (A has full column rank)

  <u>Fact:</u> when $\text{rank}(A) = N$, then for each $\underline{b} \in R(A)$ there exists a unique $\underline{x} \in \mathbb{C}^N$ s.t. $A\underline{x} = \underline{b}$

  Due to uniqueness, $\exists \mathcal{A}^L : R(A) \to \mathbb{C}^N$ s.t. $\mathcal{A}^L(\underline{b}) = \underline{x}$

- Theorem : Existence of a Linear - Left Inverse

If $A \in \mathbb{C}^{M \times N}$ with $\text{rank}(A) = N$, then $\exists A^L \in \mathbb{C}^{M \times N}$ s.t.

$$\left.\begin{array}{ll} \text{(i)} & A(A^L \underline{b}) = \underline{b} \\ \text{(ii)} & A^L(A\underline{x}) = \underline{x} \end{array}\right\} \begin{array}{l} \text{for } \forall \underline{x} \in \mathbb{C}^N \\ \text{for } \forall \underline{b} \in R(A) \end{array}$$

<u>Proof :</u> $A^L(A\underline{x}) = \underline{x}$ by definition

and $A(A^L \underline{b}) = \underline{b}$ for $\forall \underline{x} \in \mathbb{C}^N$ and $\forall b \in R(A)$

Let $A\underline{x}_1 = \underline{b}_1$ and $A\underline{x}_2 = \underline{b}_2$

$$\Downarrow$$

$$\underline{x}_1 = A^L \underline{b}_1 \quad \text{and} \quad \underline{x}_2 = A^L \underline{b}_2$$

$$A(\alpha \underline{x}_1 + \beta \underline{x}_2) = \alpha \underline{b}_1 + \beta \underline{b}_2$$

$$A^L(\alpha \underline{b}_1 + \beta \underline{b}_2) = \alpha \underline{x}_1 + \beta \underline{x}_2 = \alpha A^L(b_1) + \beta A^L(\underline{b}_2)$$

$\Rightarrow A^L$ is linear $\Rightarrow$ we have a matrix representation for it

<u>Note:</u> (1) "left - inverse" $A^L A = I_N$

(2) We don't (yet) know how to find A^L

(3) left - inverse is not necessarily unique